to the distortion of a picture by a curved lens or mirror, as in an amusement park's fun house, for example.

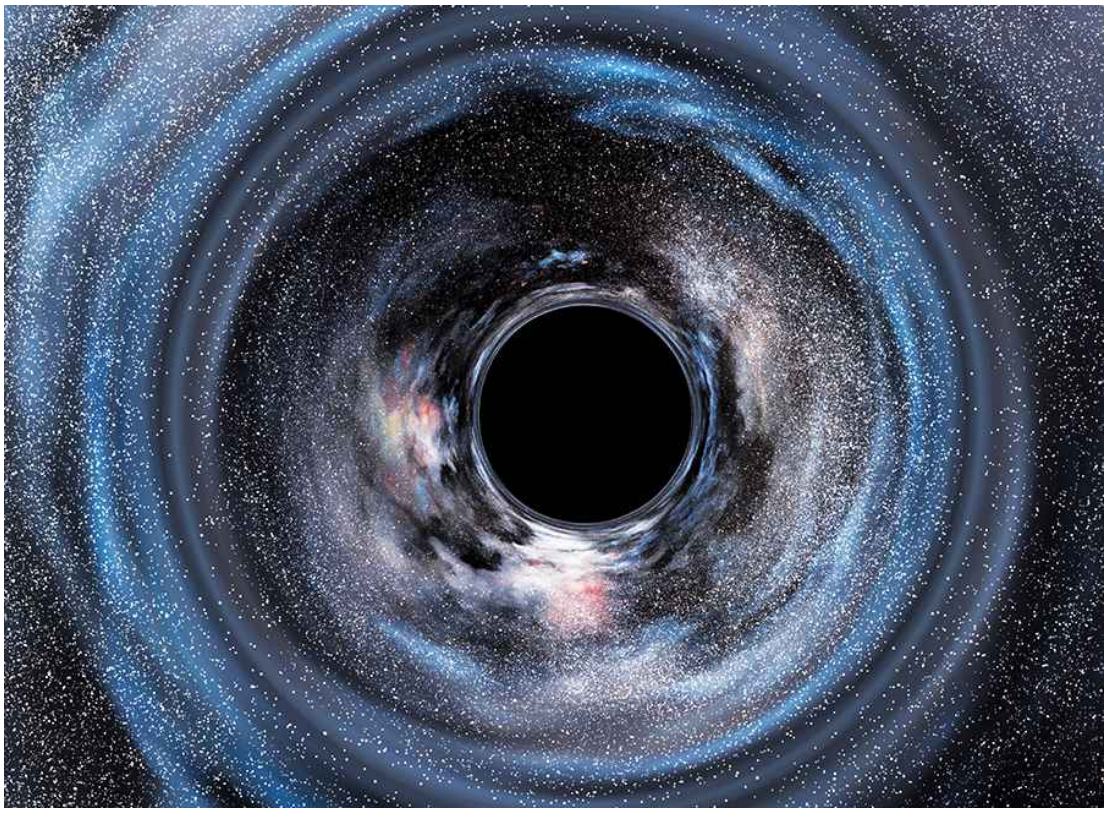


Fig. 3.3. The stars in Gargantua's galaxy, as seen around Gargantua's shadow. Gargantua bends the light rays coming from each star, thereby distorting enormously the appearance of its galaxy: "gravitationally lensing" the galaxy. *[From a simulation for this book by the Double Negative visual-effects team.]*

Einstein's relativistic laws predict, unequivocally, all the properties of black holes from their surfaces outward, including their gravitational lensing.[7] Astronomers have firm observational evidence that black holes exist in our universe, including gigantic black holes like Gargantua. Astronomers have seen gravitational lensing by other objects (for example, Figure 24.3), though not yet by black holes, and the observed lensing is in precise accord with the predictions of Einstein's relativistic laws. This is enough for me. Gargantua's gravitational lensing, as